llmXive Automated Review of Learning from the Self-future: On-policy Self-distillation for dLLMs

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the support provided by citations and data within the manuscript.

The paper makes several strong claims regarding the novelty and performance of the proposed d-OPSD framework. The claim that d-OPSD is the "first OPSD framework tailored for dLLMs" (Abstract, Section 3.3) is supported by a clear distinction from existing works like d3LLM and Cd4lm, which are characterized as off-policy or SFT-like. The citations provided (qian2026d3llm, liang2026cd4lm) appear to support this distinction in the text, though the reader must accept the authors' definition of "on-policy" in the diffusion context as the differentiating factor.

The performance claims in Section 4.2 and Table 3 state that d-OPSD requires "only around 10% of the optimization steps" compared to RLVR. The data in Table 3 supports a significant reduction, but the specific percentages vary by task (e.g., GSM8K: 425/7700 ≈ 5.5%; Sudoku: 425/3800 ≈ 11.2%). While "around 10%" is a reasonable summary, the phrasing could be slightly more precise to reflect the range observed in the data, ensuring the claim is not perceived as an exact average when the variance is notable.

In Section 4.3, the authors claim that the AR-style OPSD baseline "fails to bring new knowledge" based on an "Overlap Top-K" metric approaching 1.0. While the metric is defined, the claim of "failure" relies on the implicit assumption that an overlap of ~1.0 indicates no new information transfer. The text would benefit from explicitly stating the theoretical maximum overlap or a baseline (e.g., random chance) to contextualize why a value near 1.0 is detrimental, thereby strengthening the causal link between the metric and the claim of "no new knowledge."

Overall, the claims are generally well-supported by the provided tables and figures, but minor clarifications on the interpretation of specific metrics and the precision of summary statistics would enhance the accuracy and robustness of the assertions.

Action items.

- [writing] In Section 4.3, the claim that AR-style OPSD 'fails to bring new knowledge' relies on an 'Overlap Top-K' metric near 1.0. Explicitly state the theoretical baseline or random overlap value to contextualize why this metric implies a lack of new information transfer.
- [writing] Section 4.2 claims d-OPSD uses 'around 10%' of RLVR steps. Table 3 shows a range from ~5.5% (GSM8K) to ~11% (Sudoku). Refine the phrasing to '5-11%' or 'approximately 10%' to accurately reflect the data variance across tasks.
- [writing] The claim of being the 'first OPSD framework for dLLMs' (Abstract, Sec 3.3) distinguishes d-OPSD from d3LLM and Cd4lm based on 'on-policy' vs 'off-policy' nature. Ensure the citations clearly support this specific distinction and that the definition of 'on-policy' excludes the cited methods' approaches.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear visual overview of the d-OPSD framework, but the caption filename is incorrect, and the distinction between the problem statement and the reference solution in the 'Teacher Prompt' box is not explicitly defined in the legend.

Figure 2

Figure 2 clearly displays a sample math problem from the GSM8K training set, consistent with its caption. The text is legible and the figure serves its purpose as a qualitative example without any missing elements or misleading information.

Figure 3

Figure 3 displays a clear, step-by-step mathematical derivation and final answer for a GSM8K problem, consistent with its caption 'The self-generated future answer.' The text is legible, the logic is transparent, and there are no missing labels or misleading elements.

Figure 4

The figure is a static screenshot of a text generation interface that fails to visually represent the dynamic 'decoding status' or the specific self-distillation mechanism described in the caption, rendering it ineffective for scientific communication.

Figure 5

The figure illustrates the self-teacher construction process but contains a typo in the title and suffers from poor readability due to excessive masking and highlighting.

Figure 6

The figure content is a static text example of a math problem and solution, which does not visually represent the 'AR-style teacher construction' process described in the caption.

Figure 7

Figure 7 displays a clear text-based math problem from the GSM8K dataset, consistent with its caption. The image is legible and effectively serves as a qualitative example of the training data.

Figure 8

The figure displays a raw model generation with unrendered formatting tags and delimiters, which reduces its readability and professional appearance. The caption is functional but lacks context linking it to the specific problem instance.

Figure 9

The figure contains a spelling error in the title and displays text that is largely illegible due to excessive masking and highlighting, hindering the ability to verify the experiment's content.

Figure 10

The figure presents a quantitative plot of accuracy decline but fails to support the caption's claim of a 'failure mode' due to a lack of experimental context, missing baselines, and a generic title that does not match the specific phenomenon described.

Figure 11

The figure provides a useful qualitative comparison of model reasoning, but the caption is vague and the text rendering includes raw LaTeX commands that hinder readability.
Action items.

- [writing] Figure 1: The caption filename '[fig2.png]' contradicts the label 'Figure 1' and the content (which matches the description of Figure 2 in the provided list); verify the correct filename.
- [writing] Figure 1: The 'Teacher Prompt' box contains red text ('Here is a reference solution...') that is not defined in the legend or caption, making the distinction between the problem statement and the reference solution unclear.
- [science] Figure 1: The 'On-policy Generation' section shows 'Self-generated Future' tokens (e.g., 'bus', 'stop') appearing in the 'Teacher Construction' row, but the caption does not explain how the teacher is constructed from the student's future, creating a logical gap in the visual workflow.
- [writing] Figure 4: The image displays a UI screenshot titled 'Student Decoding' containing a reasoning block and masked tokens, but lacks the visual elements (e.g., timeline, divergence metrics, or state visualization) implied by the caption 'Current student decoding status' to effectively communicate the method's mechanics.

- [science] Figure 4: The figure shows a static text snippet with masked tokens (◁mask▷) but does not visually demonstrate the 'decoding status' or the specific 'self-future' information being utilized, making it difficult to verify the claim of on-policy self-distillation.
- [writing] Figure 5: The title contains a typo ('Constructiton' instead of 'Construction').
- [writing] Figure 5: The text content is heavily obscured by red highlighting and ◁mask▷ tokens, making the specific construction steps at $t = 20$ difficult to read.
- [science] Figure 6: The image displays a static math problem and reference solution (GSM8K style) rather than a diagram or visualization of 'AR-style teacher construction' as claimed by the caption. It fails to illustrate the autoregressive process or mechanism described.
- [science] Figure 8: The image displays raw model output with XML tags (e.g., <reasoning>, </reasoning>) and LaTeX delimiters (e.g.,

  $$\ldots$$

) that are not rendered or stripped. This makes the figure look like a raw log rather than a polished 'self-generated answer' for a scientific paper.
- [writing] Figure 8: The caption 'The self-generated answer' is too brief. It should explicitly state that this is the output corresponding to the question in Figure 7 to provide necessary context.
- [writing] Figure 9: The title contains a spelling error ('Construciton' instead of 'Construction').
- [science] Figure 9: The text content is heavily obscured by ◁mask▷ tokens and red highlights, making the specific 'toy experiment' data illegible and preventing verification of the self-teacher construction.
- [science] Figure 10: The caption 'Failure Mode of collapse' describes a qualitative failure, but the figure displays a quantitative line plot of accuracy vs. steps without context. It is unclear what task, dataset, or baseline this represents, making the claim of 'collapse' unsubstantiated by the visual alone.
- [writing] Figure 10: The title 'Countdown' is generic and does not reflect the caption's specific claim of 'Failure Mode of collapse'; the title should be descriptive of the phenomenon shown.
- [science] Figure 10: The x-axis label 'Optimization Steps' and the sharp drop at step 275 suggest a specific experimental condition, but the caption provides no details on the setup, making the figure impossible to interpret or reproduce.
- [writing] Figure 11: The caption 'Qualitative Examples on GSM8k' is too generic; it should explicitly describe the content (e.g., 'Comparison of d-OPSD vs. RLVR reasoning traces on a GSM8K math problem').

- [writing] Figure 11: The raw LaTeX code (e.g., `\text{}`, `\boxed{}`) is visible in the reasoning traces, reducing readability and suggesting incomplete rendering.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on domain-specific acronyms and jargon that are not consistently defined for a broader audience. In the Abstract, the terms “dLLMs” and “OPSD” are introduced and used immediately without defining the full phrases “diffusion Large Language Models” and “On-policy Self-distillation” first. While “dLLMs” is defined in the Introduction, the Abstract should be self-contained. Similarly, “RLVR” is used in the Introduction without its full expansion (“Reinforcement Learning with Verifiable Rewards”) being provided at the point of first mention.

There are several instances where jargon or technical phrasing is marred by grammatical errors that reduce clarity. In Section 3.2, the phrase “nature choice” should be “natural choice.” In Section 3.3, “severing simultaneously” is a typo for “serving simultaneously.” In Appendix 1.2, “spare rewards” is likely a typo for “sparse rewards,” which is a critical technical distinction in reinforcement learning contexts. Additionally, the phrase “fix teacher strategy” in Section 3.3 should be “fixed teacher strategy” to maintain grammatical parallelism.

The term “self future-experience” is used repeatedly (Abstract, Introduction, Section 3.1, Conclusion) as a branded concept. While creative, it is not standard terminology and is used without a formal definition or explanation of its precise mathematical or operational meaning beyond the metaphor of “daydreaming.” A clearer, more standard definition would improve accessibility. Finally, the phrase “block-diffusion” in Appendix 1.1 is introduced as a “common-used inference strategy” but lacks a concise definition for readers unfamiliar with the specific hybrid AR/diffusion mechanics described.

Action items.

- [writing] The manuscript relies heavily on domain-specific acronyms and jargon that are not consistently defined for a broader audience. In the Abstract, the terms “dLLMs” and “OPSD” are introduced and used immediately without defining the full phrases “diffusion Large Language Models” and “On-policy Self-distillation” first. While “dLLMs” is defined in the Introduction, the Abstract should be self-contained. Similarly, “RLVR” is used in the Introduction without its full expansion (“Reinforcement Learning

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper is generally strong in its high-level motivation: adapting On-Policy Self-Distillation (OPSD) to Diffusion LLMs (dLLMs) requires changing both the conditioning mechanism (prefix to suffix) and the supervision granularity (token to step). The argument that AR-style prefix conditioning is incompatible with dLLMs' arbitrary-order generation is well-founded.

However, there are specific logical gaps in the mathematical formulation and the causal explanation of the results:

1. Ambiguity in Step-Level Divergence (Section 3.2): The definition of the subset $\mathcal{K}_t$ (Eq 11) is logically opaque. The text describes selecting the "top-k most confident tokens," but the equation $\sum |\mathcal{K}_t| = L$ suggests a partitioning of the sequence length. It is unclear if k is a fixed hyperparameter per step or if the set size varies. Furthermore, Eq 12 averages the KL divergence over $\mathcal{K}_t$. If $\mathcal{K}_t$ is selected based on confidence, the supervision is biased toward high-confidence tokens. The paper does not logically justify why this specific bias is beneficial for dLLMs compared to a uniform step-level average, nor does it clearly define the selection algorithm.
2. Causal Link in "New Knowledge" Claim (Section 4.3): The paper argues that the AR-style baseline fails because the "Overlap Top-K_t" is near 1, implying no new knowledge transfer. This is a correlation presented as a causal mechanism. High overlap could simply indicate that both the student and teacher (even with different conditioning) agree on the most probable tokens for a given task. The paper fails to provide a logical argument for *why* suffix conditioning (self-future) structurally necessitates a lower overlap or higher information gain compared to prefix conditioning in the specific context of diffusion denoising. Without this mechanism, the claim that the performance gap is *due* to "new knowledge" rather than just better alignment with the diffusion process is weak.
3. The "Self-Future" Paradox (Section 3.3 & 4.4.4): The method relies on the student generating a trajectory, and the teacher being conditioned on the *final* answer of that trajectory (Eq 9). In the early stages of training, the student's final answer is likely incorrect. Logically, conditioning the teacher on an incorrect "future" should guide the student toward that incorrect answer, reinforcing errors. The paper mentions a "Fix teacher" strategy (App 5.1) and a "Compute only on Correct Generations" strategy (App 5.2), but the core method description in Section 3.3 does not explicitly state that the teacher construction *requires* a correct final answer to be valid. If the method is applied to incorrect trajectories, the logical consistency of "learning from self-future" breaks

down. The paper needs to clarify if the "on-policy" loop is strictly gated by correctness, which would make it a form of filtered on-policy learning rather than pure on-policy.

Action items.

- [writing] Ambiguity in Step-Level Divergence (Section 3.2): The definition of the subset $\mathcal{K}_t$ (Eq 11) is logically opaque. The text describes selecting the "top-k most confident tokens," but the equation $\sum |\mathcal{K}_t| = L$ suggests a partitioning of the sequence length. It is unclear if k is a fixed hyperparameter per step or if the set size varies. Furthermore, Eq 12 averages the KL divergence over $\mathcal{K}_t$. If $\mathcal{K}_t$ is selected based on confidence, the supervision
- [writing] Causal Link in "New Knowledge" Claim (Section 4.3): The paper argues that the AR-style baseline fails because the "Overlap Top-K_t" is near 1, implying no new knowledge transfer. This is a correlation presented as a causal mechanism. High overlap could simply indicate that both the student and teacher (even with different conditioning) agree on the most probable tokens for a given task. The paper fails to provide a logical argument for *why* suffix conditioning (self-future) structurally neces
- [writing] The "Self-Future" Paradox (Section 3.3 & 4.4.4): The method relies on the student generating a trajectory, and the teacher being conditioned on the *final* answer of that trajectory (Eq 9). In the early stages of training, the student's final answer is likely incorrect. Logically, conditioning the teacher on an incorrect "future" should guide the student toward that incorrect answer, reinforcing errors. The paper mentions a "Fix teacher" strategy (App 5.1) and a "Compute only on Correct Generati

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a compelling adaptation of On-Policy Self-Distillation (OPSD) for Diffusion Large Language Models (dLLMs), introducing a novel "self-future" conditioning mechanism. However, the manuscript occasionally overreaches in its claims regarding efficiency and the exclusivity of its approach.

First, the Abstract and Section 4.2 claim that d-OPSD requires "only around 10% of the optimization steps" compared to RLVR. While Table 3 supports this regarding the number of gradient updates, the text explicitly notes in Section 3.3 and 4.2 that d-OPSD employs a pass@k sampling strategy (k=8) to ensure correct trajectories, which shares the "same computation overhead as RLVR (group k rollouts) for each training step." Consequently, the total computational cost (FLOPs) or wall-clock time may not be 10% of the baseline, even if the number of updates is. The claim of "superior sample efficiency" is technically correct in terms of updates but risks misleading readers into assuming a 10x reduction in total compute. The text should clarify that the efficiency gain is specific to the number of optimization steps, not necessarily the total computational budget.

Second, the Introduction and Abstract characterize existing OPSD methods as "inherently autoregressive-centric" and "fundamentally conflicting" with dLLMs. This language is too absolute. The paper demonstrates that the proposed suffix-conditioning approach outperforms an AR-style prefix-conditioning baseline (Table 4), but it does not prove that the AR-style approach is theoretically impossible or universally ineffective for dLLMs. The conflict is empirical and performance-based, not fundamental. The authors should soften this to state that the AR-style approach is "suboptimal" or "less effective" rather than "fundamentally conflicting," which implies a theoretical impossibility that is not fully substantiated.

Finally, the claim of being the "first OPSD framework tailored for dLLMs" relies heavily on the distinction between using self-generated trajectories (on-policy) versus static ground-truth data (off-policy, as in d3LLM). While this distinction is valid, the manuscript would benefit from a more explicit definition of "on-policy" in the context of self-distillation early on to ensure the "first" claim is not misinterpreted as ignoring all prior self-distillation work in diffusion models. The current phrasing is slightly ambiguous regarding whether previous works attempted on-policy updates with different conditioning strategies.

Action items.

- [writing] The claim of '10% of optimization steps' (Abstract) conflates update count with compute. Since d-OPSD uses pass@8 sampling per step (Sec 3.3), total FLOPs may match RLVR. Clarify that efficiency is in updates, not total compute.
- [writing] Describing AR-style OPSD as 'fundamentally conflicting' (Abstract) overstates the case. Table 4 shows it is less effective, not impossible. Soften to 'suboptimal' to avoid implying theoretical impossibility.
- [writing] The 'first OPSD for dLLMs' claim (Abstract) relies on distinguishing on-policy vs. off-policy data sources. Explicitly define this distinction early to prevent ambiguity regarding prior self-distillation attempts in diffusion models.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a novel framework, d-OPSD, for on-policy self-distillation in diffusion large language models (dLLMs). From a safety and ethics perspective, the primary concerns revolve around the potential for unintended consequences of the self-distillation process, the transparency of the training methodology, and the broader implications of the technique.

First, the methodology relies on a “self-teacher” constructed from the model’s own correct generations (Pass@k). As noted in Appendix A.4.2, the authors implement a mechanism to clear unmasked tokens in new blocks to prevent “leaking the final answer.” While this is a technical necessity for the distillation process, it raises questions about the integrity of the training signal. If the model fails to generate a correct answer, the sample is discarded or the process is repeated. This filtering could introduce a bias towards specific reasoning patterns that are more likely to be correct, potentially leading to a model that is less robust to edge cases or adversarial inputs. The paper should clarify the ethical implications of this filtering process and whether it could inadvertently reinforce specific reasoning paths while discarding others, which might lead to model collapse or reduced robustness.

Second, the paper acknowledges “policy collapse” as a failure mode in Section 4.5, where the model’s performance degrades catastrophically after reaching a peak. While this is a technical limitation, the authors should discuss the safety implications of deploying a model trained with d-OPSD that is prone to collapse. If the model collapses during inference or fine-tuning, could it generate harmful or nonsensical outputs more frequently than baseline models? A brief discussion on the safety risks associated with this instability is recommended, particularly in the context of deploying such models in real-world applications.

Third, while the current experiments are conducted on benign datasets like GSM8K and MATH500, the methodology (d-OPSD) is general and could be applied to other domains. The authors should include a statement in the ‘Limitations’ or ‘Ethical Considerations’ section acknowledging that while their current experiments are on benign tasks, the method could theoretically be applied to generate harmful content if used with malicious intent. They should also clarify that they have not evaluated the safety of the generated outputs beyond correctness, and that future work should include safety evaluations to ensure the model does not generate harmful content.

Overall, the paper is well-written and the methodology is sound, but a more thorough discussion of the safety and ethical implications of the proposed technique is needed to ensure that the research is conducted responsibly.

Action items.

- [writing] The paper presents a novel framework, d-OPSD, for on-policy self-distillation in diffusion large language models (dLLMs). From a safety and ethics perspective, the primary concerns revolve around the potential for unintended consequences of the self-distillation process, the transparency of the training methodology, and the broader implications of the technique. First, the methodology relies on a “self-teacher” constructed from the model’s own correct generations (Pass@k). As noted in Appendix A

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a novel adaptation of On-Policy Self-Distillation (OPSD) for Diffusion Large Language Models (dLLMs), proposing d-OPSD. The core scientific claims regarding the method's design (suffix conditioning, step-level divergence) are logically derived from the properties of dLLMs. However, the strength of the evidence supporting the central claims of "superior sample efficiency" and the robustness of the ablation findings requires clarification.

First, the claim of superior sample efficiency (Abstract; Section 4.2) states that d-OPSD requires "only around 10% of the optimization steps" compared to RLVR. While Table 3 lists the number of gradient updates (425 vs 7700 for GSM8K), the paper notes in Section 3.3 that d-OPSD employs a pass@k sampling strategy (k=8) to find a correct trajectory before updating, which shares the "computation overhead" of RLVR's group rollouts. The term "sample efficiency" in RL contexts typically refers to the number of environment interactions or total compute required to reach a performance threshold. If d-OPSD requires 8x more forward passes per update to find a correct trajectory, the total compute cost might be comparable to RLVR, even if the number of *updates* is lower. The manuscript conflates "optimization steps" with "sample efficiency" without normalizing for the total computational cost or wall-clock time. To support the claim of efficiency, the authors must provide a comparison based on total FLOPs or wall-clock time, or explicitly redefine the metric to avoid ambiguity.

Second, the ablation study on the retaining ratio ρ_{teacher} (Table 6, Section 4.4) presents a counter-intuitive result where a weaker teacher ($\rho = 0.10$, 80.5% accuracy) outperforms a stronger teacher ($\rho = 0.50$, 79.8% accuracy). While the authors hypothesize that a weaker teacher provides a better learning signal, the difference is marginal (0.7%) and lacks statistical validation. Given the stochasticity of LLM training, this could be noise. The paper does not report standard deviations, confidence intervals, or results from multiple random seeds for these ablation runs. Without this statistical rigor, the conclusion that "distillation effectiveness is not only decided by the teacher performance" is not fully supported by the evidence provided.

Finally, the discussion of failure modes (Section 4.5) attributes policy collapse to "model-seeking behavior" becoming overly narrow. This is a plausible hypothesis consistent with prior RL literature, but the paper offers no quantitative evidence to substantiate it. There are no plots showing the evolution of the KL divergence, policy entropy, or the distribution of generated tokens over the training steps where collapse occurs. Without such data, the explanation remains speculative. The authors should include diagnostic plots or metrics to demonstrate that the collapse correlates with the predicted narrowing of the policy distribution.

In summary, while the methodological contribution is sound, the evidence for the specific claims of efficiency and the interpretation of ablation results needs to be strengthened with more rigorous statistical analysis and compute-normalized comparisons.
Action items.

- [science] The 'sample efficiency' claim compares optimization steps but ignores that d-OPSD uses pass@8 sampling per step. Without normalizing by total FLOPs or wall-clock time, the claim is ambiguous and potentially misleading.
- [science] The ablation on retaining ratio (Table 6) shows a weaker teacher outperforming a stronger one. This counter-intuitive finding lacks statistical validation (e.g., error bars, multiple seeds) to distinguish signal from noise.
- [science] The failure mode analysis attributes collapse to 'model-seeking behavior' but provides no quantitative evidence (e.g., KL/entropy trajectories) to support this hypothesis over other causes like overfitting.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a novel framework, d-OPSD, for on-policy self-distillation in diffusion large language models. While the methodological adaptation is sound, the statistical rigor of the experimental evaluation is insufficient for a top-tier venue.

Lack of Variance Reporting: The primary statistical deficiency is the absence of variance estimates for all quantitative results. Tables 1 through 8 report single-point accuracy scores (e.g., "81.0" for GSM8K in Table 2) without standard deviations, standard errors, or confidence intervals. In deep learning experiments, performance can vary significantly across random seeds due to initialization and stochastic optimization. Without reporting results averaged over multiple seeds (e.g., $N \geq 3$) with error bars, it is impossible to determine if the reported improvements (e.g., the 1.2% gain over diffu-GRPO in Table 2) are statistically significant or merely artifacts of a specific random seed. The claim of "consistent outperformance" in Section 4.2 is not statistically supported by single-point estimates.

Missing Significance Testing: The paper claims d-OPSD "consistently outperforms" baselines (Section 4.2) and that specific ablation choices (e.g., $\rho_{\text{teacher}} = 0.25$ vs 0.10 in Table 6) yield better results. However, no statistical hypothesis tests (e.g., paired t-tests or Wilcoxon signed-rank tests) are performed to validate these claims. Given the small number of tasks (4) and the likely high variance in LLM reasoning benchmarks, formal testing is required to distinguish signal from noise.

Ablation Study Rigor: The ablation studies in Section 4.4 (Tables 5-8) are critical for validating the design choices (e.g., step-level vs. token-level, retaining ratios). Currently, these tables present single runs. The differences observed (e.g., 81.0 vs 79.8 in Table 6) are small and may not be robust. The authors must re-run these ablations with multiple seeds and report the mean ± standard deviation to confirm that the chosen hyperparameters are genuinely optimal and not overfit to a specific run.

Sample Efficiency Claims: The claim of "superior sample efficiency" (Section 4.2, Table 3) relies on comparing convergence steps. While the step counts are distinct (425 vs 7700), the learning curves (Figure 1) lack error bands. To robustly support the efficiency claim, the learning curves should show performance variance over time, demonstrating that the faster convergence is consistent and not a fluke of the specific training trajectory.

In summary, the paper requires a statistical overhaul of its experimental section. The authors should re-run experiments with multiple seeds, report mean ± standard deviation for all metrics, and apply appropriate statistical tests to validate their comparative claims.
Action items.

- [science] Report confidence intervals or standard deviations for all reported accuracy metrics in Tables 1-8. Single-point estimates without variance measures (e.g., from multiple seeds) make it impossible to assess statistical significance of the claimed improvements over baselines.
- [science] Clarify the statistical test used to claim 'consistent outperformance' in Section 4.2. With only 4 tasks and varying sequence lengths, a formal test (e.g., paired t-test or Wilcoxon signed-rank) across seeds is required to support the claim that d-OPSD is statistically superior to diffu-GRPO.
- [science] The ablation studies in Section 4.4 (Tables 5-8) present single-point results. Provide variance estimates (e.g., standard error) for these ablation results to determine if observed differences (e.g., 81.0 vs 80.5) are statistically significant or within noise.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a novel framework for on-policy self-distillation in diffusion large language models (dLLMs). The core ideas are innovative, and the experimental results appear compelling. However, the writing quality requires minor revisions to address several grammatical errors, subject-verb agreement issues, and awkward phrasings that occasionally impede readability.

Specifically, in the Introduction (Section 1), the sentence "While recent works... has successfully applied" contains a subject-verb agreement error; "works" is plural and requires "have." In Section 3.2, the phrase "a nature choice" is incorrect and should be "a natural choice." Section 3.3 contains a significant typo where "severing simultaneously" is used instead of "serving simultaneously," which alters the meaning.

In the Experiments section, Section 4.4 includes the phrase "despite it is a weaker teacher," which is grammatically incorrect; it should be rephrased as "despite being a weaker teacher." Section 4.5, which discusses failure modes, has multiple issues: "training sometimes degrade" should be "degrades," "the same phenomena is" should be "the same phenomenon is," and "prevent from further learning" is missing a subject or gerund, likely intended as "preventing it from further learning."

In the Appendix, Section 3.2 uses "a engineering technique" instead of "an engineering technique." Additionally, the sentence "the gradient remains constant... and no longer needs to be stored" has a slightly ambiguous subject-verb agreement regarding "gradient maps" versus "gradient," which should be clarified. Finally, Appendix 3.3 contains "set an threshold" (should be "a threshold") and "should be include" (should be "included").

Addressing these specific grammatical and syntactical errors will significantly improve the clarity and professionalism of the manuscript without altering the scientific content.
Action items.

- [writing] Correct subject-verb agreement in Section 1: 'While recent works... has successfully applied' should be 'have successfully applied'.
- [writing] Fix article usage and phrasing in Section 3.2: 'a nature choice' should be 'a natural choice'.
- [writing] Correct grammatical error in Section 3.3: 'severing simultaneously' should be 'serving simultaneously'.
- [writing] Fix sentence structure in Section 4.4: 'despite it is a weaker teacher' should be 'despite being a weaker teacher' or 'despite it being a weaker teacher'.
- [writing] Correct subject-verb agreement in Section 4.5: 'training sometimes degrade' should be 'training sometimes degrades'.
- [writing] Fix article usage in Section 4.5: 'the same phenomena is' should be 'the same phenomenon is'.
- [writing] Correct preposition usage in Section 4.5: 'prevent from further learning' should be 'preventing it from further learning' or 'preventing further learning'.
- [writing] Fix article usage in Appendix 3.2: 'a engineering technique' should be 'an engineering technique'.
- [writing] Correct verb form in Appendix 3.2: 'need to be stored' should be 'needs to be stored' to agree with 'gradient maps' or rephrase for clarity.
- [writing] Fix article usage in Appendix 3.3: 'set an threshold' should be 'set a threshold'.
- [writing] Correct verb form in Appendix 3.3: 'should be include' should be 'should be included'.